Abhay Tiwari

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Education

Indian Institute of Information Technology, Raichur, Bachelor of Technology in AI and DS — Sept 2023 – Expected May 2027

- GPA: 7.9/10.0
- **Coursework:** Data Structures and Algorithms, Artificial Intelligence, Data Science, Computer Networks, Operating Systems, Object-Oriented Programming, Database Management Systems, Digital Image Processing

Experience

Indian Institute of Technology Hyderabad, Research Intern — Jan 2026 – Present

- Working under **Dr. R. Maheswaran** on temporal and spatial downscaling of GRACE satellite data.
- Engineered a high-precision Residual U-Net CVAE for multi-channel geophysical data. The model outperformed traditional baselines through custom spatial regularization, reaching a correlation of 0.98 and a Nash-Sutcliffe Efficiency (NSE) of 0.94.

Projects

Agentic Legal Assistant (Law-Agentic System) | *LangGraph, LangChain, FastAPI, Firebase, Google Gemini* — Github Repository

- Architected a multi-role **Agentic AI** system using **LangGraph** to automate legal workflows (filing, assignment, reporting) for Complainers, Inspectors, and Prosecutors.
- Implemented a **Retrieval-Augmented Generation (RAG)** pipeline using **FAISS** and **HuggingFace Embeddings** to ground responses in Indian Legal Codes (BNS, IPC, CrPC).
- Deployed the agent via a **FastAPI** backend with streaming capabilities and integrated **Firebase Firestore** to perform real-world database transactions like filing complaints and assigning cases.

Reinforcement Learning for Lunar Landing | *PyTorch, Gymnasium, Stable Baselines3* — Github Repository

- Developed a Deep Reinforcement Learning agent to autonomously solve the LunarLander-v3 environment using Python and the Stable Baselines3 library.
- Implemented and fine-tuned a Proximal Policy Optimization (PPO) algorithm with 16 parallel environments to accelerate learning and achieve stable flight control.

Deep Learning Solutions for Biharmonic Equations| *Python, PyTorch, CUDA, PINNs, Deep Ritz Method* — Github Repository

- Engineered a comparative deep learning framework to solve fourth-order Biharmonic Partial Differential Equations (PDEs) commonly used in elasticity and fluid mechanics.
- Implemented two distinct solvers: a Physics-Informed Neural Network (PINN) handling the strong PDE form via automatic differentiation, and a Deep Ritz Method (DRM) model minimizing the variational potential energy.
- Achieved high-fidelity results with 1.4% relative error using PINNs, while demonstrating superior scalability with DRM by attaining a 10x reduction in computational time via GPU acceleration.

Technical Skills

Languages: Python, C++, C, Java, Go, SQL

Technologies: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, PyTorch, LangChain, LangGraph, Physics-Informed Neural Networks (PINNs), Deep Ritz Method, Django, FastAPI

Tools & Platforms: CUDA, Git, SQLite, PostgreSQL, Google Colab, Jupyter Notebook, VS Code

Coding Profile: Leetcode: Solved 200+ Problems (Link), Top 120 Rank - Amazon ML Challenge 25